

OneModel Traffic Light Annotation Specification

Chery — Automotive AI Division

Original: OneModel/■■■■■■■■■ | Version: V-E2E-2.1

Last Updated: April 17, 2026

PARTIAL CONTENT: Sections 1 (Overview & Abnormal Images) are fully translated. Section 2 (TL Rules — Lamp Box, Lamp Bulb, Countdown) structure is present but content requires additional Feishu page access. This PDF will be updated when full content is available.

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1. Version / Change Log

Date	Version	Changes	Changed By	Proposer	Confirmer
2026-03-17	V-E2E-2.0	Merged end-to-end traffic light & TL traffic rules E2E changes: 1. Added bulb shapes: Prohibition Lamp, Passage Lamp 2. Changed invalid attribute definition 3. Flashing attribute: flashing starts from first re-illumination after blackout TL changes: 1. Added ignore-frame attribute 2. Adjusted occlusion ratio, merged truncation & occlusion 3. Key-point requirement changed: unlit bulbs also need key points 4. No longer judge guided-straight attribute 5. Added color option: black	Yi Tao (■)	Yi Tao (■)	Yi Tao (■)
2026-03-27	V-E2E-2.1	TL wide-angle starting pixel adjusted: use shoulder > 5px as starting point	Yi Tao (■)	Yi Tao (■)	Yi Tao (■)

2. Annotation Overview

2.1 Annotation Description

This specification covers **2 traffic light annotation projects**:

- **TL (Bounding Box)**: Annotate traffic light lamp boxes and lamp bulbs in images according to the rules. Lamp boxes are annotated as **bounding boxes**, lamp bulbs are annotated as **center points**.
- **Passability Classification (Quick Judgment)**: Based on the current intersection's traffic light or passage lamp for the ego vehicle's lane, determine the passability of the road.

Annotated Camera Names:

During annotation, two cameras must be used for judgment:

- **Telephoto camera (30°)**: CAM_PBQ_FRONT_TELE_RESET_OPTICAL_H30_HR — [Manual annotation data]
- **Wide-angle camera (120°)**: CAM_PBQ_FRONT_WIDE_RESET_OPTICAL_H120_HR — [Manual annotation data]

Camera overview images:

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OneModel红绿灯标注规范

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1.2.2 行人灯

1.2.3 自行车灯

1.2.4 其他 Other

1.2.5 未知 unknown

1.3 道路

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2.2 形状

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变更/版本记录

更新时间	版本	变更内容	变更人	需求提出人	需求确认人	备注
2026.3.17	V-EZE-2.0	合并针对闯红灯与TL红绿灯规则 针对闯红灯变更： 1. 新增灯泡形状：禁止通行灯、允许通行灯 2. 变更无效属性的定义 3. 闪烁属性变更，第一次灭掉后亮起的就初始给闪烁是。 TL红绿灯规则变更： 1. 新增忽略帧属性 2. 调整遮挡比例，合并截断与遮挡 3. 变更关键点要求，不亮灯泡也要打点 4. 不再判断引导直行属性 5. 颜色属性新增选项：黑色	袁艺	易铂	易铂	
2026.3.27	V-EZE-2.1	TL广角起标像素调整：按短边>5PX做起标要求	袁艺	易铂	易铂	

一、标注概述

1. 标注概述

本规则共标注2个红绿灯项目：2D拉框打点的TL规则，和快判类可通行性判断。

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2026.3.17	V-E2E-2.0	合并针对闯红灯灯与TL红绿灯规则 端对端红变更： 1. 新增灯泡形状：禁止通行灯、允许通行灯 2. 变更无效属性的定义 3. 闪烁属性变更，第一次灭掉后亮起的就开始给闪烁是。 TL红绿灯规则变更： 1. 新增忽略帧属性 2. 调整道挡比例，合并截断与道挡 3. 变更关键点要求，不亮灯泡也要打点 4. 不再判断引导直行属性 5. 颜色属性新增选项：黑色	袁艺	易铂	易铂	
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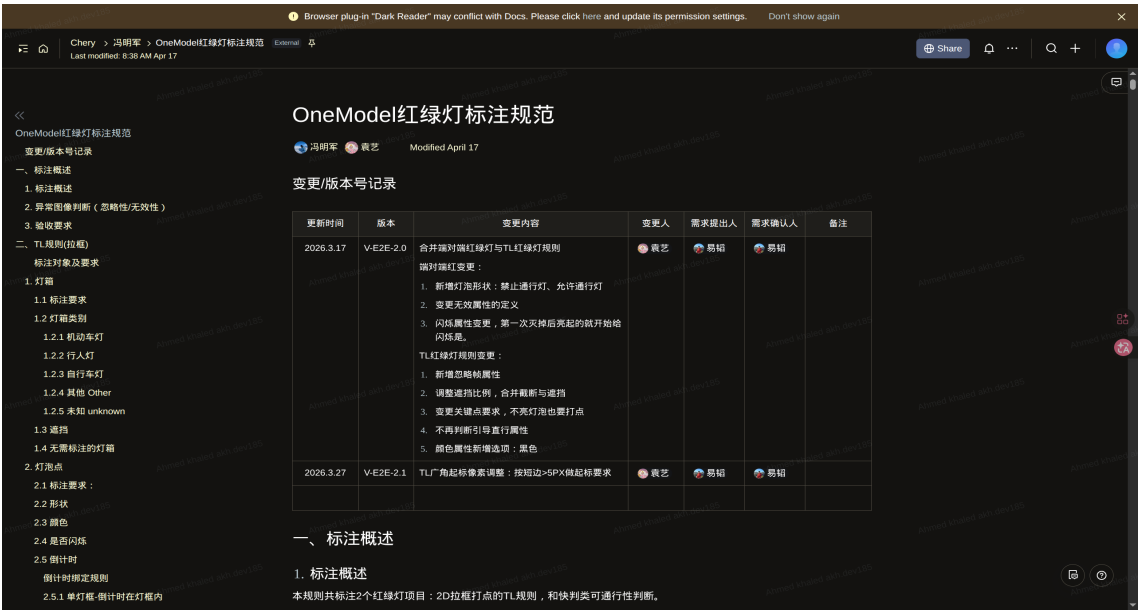
2.2 Abnormal Image Judgment (Ignorable / Invalid)

For any other 2D anomalies not listed below, provide feedback with priority.

The bounding box project and quick-judgment project need to be evaluated separately. Situations may arise where a bounding box project's non-subject lens needs passability judgment (and vice versa).

Important: The anomalies below apply when **both lenses** have the issue. If only a single lens has an anomaly, that anomalous lens simply does not need annotation.

Scenario	TL Bounding Box (Ignore Attribute)	Passability Quick Judgment (Invalid Attribute)	Image
Frame Count Missing	Ignore Frame	Invalid	[See image]
Same-source data has 200 frames, tasks with less than 150 frames	Ignore Frame	Invalid	[See image]
Dual-View Traffic Light Status Inconsistent	Ignore Frame	Invalid	[See image]
The traffic light status differs between the telephoto and wide-angle cameras.	Attribute set to "ignore frame"	Attribute set to "invalid"	



2.3 Acceptance Requirements

[Login required — content will be added after authentication.]

3. TL Rules (Bounding Box)

Annotation Objects & Requirements

[Login required — detailed TL rules content will be added after authentication.]

3.1 Lamp Box (■)

3.1.1 Annotation Requirements

[Login required — content will be added after authentication.]

3.1.2 Lamp Box Categories

3.1.2.1 Motor Vehicle Lamp (■■■■)

Traffic lights for motor vehicles — the standard red/yellow/green signals for car lanes.

[Detailed rules and images — login required.]

3.1.2.2 Pedestrian Lamp (■■■)

Traffic lights for pedestrians — walk/stop signals at crosswalks.

[Detailed rules and images — login required.]

3.1.2.3 Bicycle Lamp (■■■■)

Traffic lights for bicycles — dedicated signals for bike lanes.

[Detailed rules and images — login required.]

3.1.2.4 Other

Other types of traffic lights not falling into the above categories.

[Detailed rules and images — login required.]

3.1.2.5 Unknown

Traffic light type that cannot be determined from the image.

[Detailed rules and images — login required.]

3.1.3 Occlusion (■)

[Login required.]

3.1.4 Lamp Boxes That Do Not Need Annotation

[Login required.]

3.2 Lamp Bulb Points (■■■■)

3.2.1 Annotation Requirements

[Login required.]

3.2.2 Shape

Defines the visual shape of each lamp bulb (circle, arrow, prohibition, passage, etc.)

[Login required.]

3.2.3 Color

Color attribute for each lamp: red, yellow, green, black, etc.

[Login required.]

3.2.4 Flashing (■■■■■)

Whether the lamp bulb is flashing/blinking. Per V2.0: flashing starts from the first re-illumination after blackout.

[Login required.]

3.2.5 Countdown Timer (■■■■)

[Login required.]

Countdown Binding Rules

3.2.5.1 Single-frame: Countdown timer is inside the lamp frame

[Images — login required.]

3.2.5.2 Dual-frame: Countdown timer is to the right of the active status lamp

[Images — login required.]

3.2.5.3 Triple-frame (no right-turn): Countdown timer is to the right of the status lamp, or in the middle of status lamps (no right-turn arrow)

[Images — login required.]

3.2.5.4 Triple-frame (with right-turn): Countdown timer is in the middle between status lamps (with right-turn arrow present)

[Images — login required.]

3.2.5.5 Quad-frame: Countdown timer is between the 2nd and 3rd status lamps, or at the rightmost position

[Images — login required.]

3.2.5.6 Multi-head: Countdown timer lamp is at the bottom of the multi-head assembly

[Images — login required.]

3.3 Common Difficult Cases

3.3.1 Abnormal Lamp Display (■■■■■■■)

[Login required.]

3.3.2 Special Lamp Box Grouping (■■■■■■■■■■)

[Login required.]

4. Key Terminology (English–Chinese Glossary)

English Term	Chinese Term	Description
Lamp Box (■ ■)	■ ■	The physical housing/enclosure of a traffic light unit — annotated as a bounding box
Lamp Bulb (■ ■)	■ ■	An individual light element within a lamp box — annotated as a center point
Bounding Box (■ ■)	■ ■	Rectangular annotation drawn around the lamp box
Center Point (■ ■ ■ ■)	■ ■ ■ ■	Point annotation placed at the center of each lit or unlit bulb
Motor Vehicle Lamp	■ ■ ■ ■	Traffic light for motor vehicles (cars, trucks, etc.)
Pedestrian Lamp	■ ■ ■ ■	Traffic light for pedestrian crossings
Bicycle Lamp	■ ■ ■ ■	Traffic light for bicycle lanes
Occlusion (■ ■)	■ ■	Partial blockage of the lamp box by other objects in the scene
Truncation	■ ■	Lamp box cut off at the image boundary (merged with occlusion in V2.0)
Flashing (■ ■)	■ ■	A lamp bulb that blinks on and off
Countdown Timer (■ ■ ■ ■)	■ ■ ■ ■	Numeric display showing remaining seconds for the current signal phase
Prohibition Lamp	■ ■ ■ ■ ■ ■	Bulb shape indicating stop/prohibition (added in V2.0)
Passage Lamp	■ ■ ■ ■ ■ ■	Bulb shape indicating go/passage allowed (added in V2.0)
Ignore Frame (■ ■ ■ ■)	■ ■ ■ ■	Frame attribute for images that should be skipped (added in V2.0)
Invalid Attribute	■ ■ ■ ■ ■ ■	Image/frame marked as invalid or unusable (definition changed in V2.0)
Wide-angle Start Pixel	■ ■ ■ ■ ■ ■ ■ ■	Minimum pixel threshold for annotation — short edge > 5px (V2.1)
Dual-View Inconsistency	■ ■ ■ ■ ■ ■ ■ ■ ■ ■	When the traffic light status differs between telephoto and wide-angle cameras
Frame Count Missing	■ ■ ■ ■ ■ ■	Source data has 200 frames; tasks with <190 frames are discarded
Guided Straight	■ ■ ■ ■ ■ ■	No longer judged in V2.0
Key Point	■ ■ ■ ■	Center point on each bulb — required even for unlit bulbs (V2.0)
Passability (■ ■)	■ ■ / ■ ■ ■ ■ ■ ■	Quick-judgment classification of whether passage is allowed
Telephoto Camera (30°)	■ ■ ■ ■ ■ ■	CAM_PBQ_FRONT_TELE_RESET_OPTICAL_H30_HR — narrow FOV camera
Wide-angle Camera (120°)	■ ■ ■ ■ ■ ■	CAM_PBQ_FRONT_WIDE_RESET_OPTICAL_H120_HR — wide FOV camera